



CERTIFIED MATERIAL TEST REPORT  
CERTIFICATE OF COMPLIANCE  
CERTIFICATE # 222709  
REVISION 0

Tube Forgings of America, Inc.  
5200 N.W. Front Avenue  
Portland, OR 97210  
ISO 9001:2015 Certified  
1/30/2025

|                 |                                                                 |         |                                                                 |
|-----------------|-----------------------------------------------------------------|---------|-----------------------------------------------------------------|
| Sold To         | DNOW LP #MI04<br>1701 Gover Pkwy.<br>Mt Pleasant, MI 48858-8140 | Ship To | DNOW LP #MI04<br>1701 Gover Pkwy.<br>Mt Pleasant, MI 48858-8140 |
| Customer ID     | 34217                                                           |         |                                                                 |
| TFA Sales Order | 325840\6-1                                                      |         |                                                                 |
| Purchase Order  | 26695930                                                        |         |                                                                 |

|                  |                       |                   |               |
|------------------|-----------------------|-------------------|---------------|
| TFA Part ID      | 50012SG               | Heat Code         | A2763D 456531 |
| Description      | 6 STD LR 90 Y-52 SEGM | Grade             | Y52           |
| Quantity         | 18                    | Raw Material Heat | 538956        |
| Customer Part ID | 456531                | Process           | Y52-1A        |
|                  |                       | Mill              | TPCO          |

|                 |             |             |             |                    |          |         |     |
|-----------------|-------------|-------------|-------------|--------------------|----------|---------|-----|
| Austenitized @: | 1700° F     | Tempered @: | 1025° F     | Stress Relieved @: |          | Medium: | OIL |
| Held For:       | 60 mins/min | Held For:   | 60 mins/min | Held For:          | mins/min |         |     |

Heat Analysis Chemistry

| C    | Mn   | P     | S     | Si   | Mo   | Cr   | Ni   | Cu   | V     | Cb    | B      | Ti    | Al   | N | Co | Ca | C.E. |
|------|------|-------|-------|------|------|------|------|------|-------|-------|--------|-------|------|---|----|----|------|
| 0.14 | 1.09 | 0.012 | 0.002 | 0.32 | 0.01 | 0.10 | 0.02 | 0.04 | <0.01 | <0.01 | 0.0005 | <0.01 | 0.02 |   |    |    | 0.33 |

Product Analysis Chemistry

| C    | Mn   | P     | S     | Si   | Mo   | Cr   | Ni   | Cu   | V     | Cb    | B       | Ti    | Al   | N | Co | Ca | C.E. |
|------|------|-------|-------|------|------|------|------|------|-------|-------|---------|-------|------|---|----|----|------|
| 0.14 | 1.09 | 0.012 | 0.001 | 0.34 | 0.01 | 0.10 | 0.02 | 0.05 | <0.01 | <0.01 | <0.0010 | <0.01 | 0.02 |   |    |    | 0.33 |

Tensile Test Results

| Test | Name | Yield       | Tensile     | % Elong | R/A % | Gage Length | HBW | In-Process Test 1 Test 2 | Tensile Orientation | Test Bar Type & Size |
|------|------|-------------|-------------|---------|-------|-------------|-----|--------------------------|---------------------|----------------------|
| 1    | BASE | 57600 (psi) | 79200 (psi) | 33.60   | 73.90 | 2"          | 161 |                          | LONGITUDINAL        | FLAT (.499" Width)   |

Charpy Impact Test Results

|  | Impact Energy |   |   |     | Lateral Expansion (Mils) |   |   | % Shear |   |   |     | Test Temp. | Size (mm) | Charpy Orientation |
|--|---------------|---|---|-----|--------------------------|---|---|---------|---|---|-----|------------|-----------|--------------------|
|  | 1             | 2 | 3 | Avg | 1                        | 2 | 3 | 1       | 2 | 3 | Avg |            |           |                    |
|  |               |   |   |     |                          |   |   |         |   |   |     |            |           |                    |

Notes/Comments

1. THIS MATERIAL WAS AUSTENITIZED AT 1700F, OIL QUENCHED, AND TEMPERED AT 1025F. 2. THIS MATERIAL IS OF SEAMLESS MANUFACTURE.

1) MATERIAL MEETS MSS-SP-75 - 2019.

2) MATERIAL MEETS NACE MR-01-75/ISO 15156, REGIONS 1, 2 AND 3 - 2015 EDITION, ALSO MEETS NACE MR-0103 2015 EDITION.

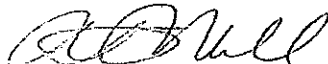
3) MANUFACTURED IN AN ISO 9001 CERTIFIED FACILITY-CERTIFICATE #30248.

4) MATERIAL UNDER 16 NPS IS NOT IMPACT TESTED UNLESS ORDERED AS Y65 OR HIGHER YIELDS OR SUPPLEMENTARY REQUIREMENT (SR24) IS INVOKED PER MSS.

\*UB = Unbroken Test

Heat: A2763D Grade: Y52

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Scot Marshall, Manager Quality Assurance

The recording of fictitious entries on this document and or its fraudulent misuse may be punishable as a felony under Federal Statute.



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|                  |                       | Mill              | TPCO   |

PRODUCT IS FIELD SEGMENTABLE.

PRODUCT MAINTAINS AN INSIDE DIAMETER NO LESS THAN 98% OF THE SPECIFIED NOMINAL INSIDE DIAMETER.

PRODUCT IS CAPABLE OF RECEIVING A PIGGING DEVICE SUITED TO TRAVEL THROUGH A FITTING WITH AN INSIDE DIAMETER NOT LESS THAN 98% OF THE SPECIFIED NOMINAL INSIDE DIAMETER.

Material Formed/Forged At: Tube Forgings of  
America, Inc, Portland OR

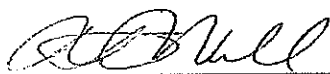
Material Heat Treated At: Met-Tek,  
Clackamas OR

TFA FITTINGS ARE MANUFACTURED IN PORTLAND, OR, USA. FOR THIS HEAT CODE, RAW MATERIAL SOURCE IS CHINA

\*UB = Unbroken Test

Heat: A2763D Grade: Y52

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